

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes/No/NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt	Yes
4	Database schema / seed files	NA (no persistent DB used)
5	Environment configuration file (.env.example)	Yes (for Watson ML credentials placeholder)
6	Deployed application URL	Pending IBM Watson ML deployment
7	APK / Executable binary	NA
8	Dockerfile / Containerization config	NA (not used in this scope)
9	Test files and test results	Yes (see Project Testing phase)
10	Demo video or walkthrough	Pending

Step 2: File / Folder Structure

```
Credit-Card-Fraud-Detection/  
├── 1. Brainstorming & Ideation/  
├── 2. Requirement Analysis/  
├── 3. Project Design Phase/  
├── 4. Project Planning Phase/  
└── 5. Project Development Phase/  
    ├── model/  
    │   ├── train_model.py  
    │   ├── deploy_watson_ml.py  
    │   ├── flask_app/  
    │   └── app.py  
    ├── templates/index.html  
    └── requirements.txt  
├── 6. Project Testing/  
├── 7. Project Documentation/  
└── 8. Project Demonstration/
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	To be added once IBM Watson ML deployment is live
Login Credentials (Demo)	NA - no auth layer implemented in current scope
Platform / Hosting Provider	IBM Watson Machine Learning (planned)
Repository Link	https://github.com/AyushaSanjuktha/credit-card-fraud-detection (update once pushed)
Demo Video Link	To be added

Step 4: Run Instructions

1. Clone the repository and cd into the 5. Project Development Phase folder.
2. pip install -r requirements.txt
3. Download creditcard.csv from <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud> and place it inside model/.
4. Run python model/train_model.py to train and save the best model (best_model.pkl, scaler.pkl).
5. cd flask_app && python app.py
6. Open <http://127.0.0.1:5000> in a browser to use the prediction form.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	IBM Watson ML deployment requires paid/trial IBM Cloud credentials not included in this repo	Run locally via Flask; deploy_watson_ml.py provided for when credentials are available
2	No authentication on the analyst dashboard	Acceptable for current academic-project scope; noted as a future enhancement

Dataset features are PCA-anonymized, limiting feature-level explainability

Model relies on aggregate statistical patterns rather than interpretable business rules